

SMITHIAN FOLD THEORY - COMPUTATIONAL PROOF PAPER 001

From Sequence to an Auditable Fold

SFT V3 Protein Fold Computational Proof - Preliminary Results

Exact representation, certified frontier search and the restart of a generalised blind-parity programme

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UPDATED PRELIMINARY-RESULTS VERSION - INVESTIGATION CONTINUES

Abstract

Protein folding is a decisive test of whether a first-principles system can generate physical, chemical and biological structure without fitted laws or target-selected rules. This first standalone paper reports the preliminary results of the Smithian Fold Theory V3 Protein Fold computational proof programme. A registered sequence and condition carrier generates a finite 24-sector torsion-word surface, a source-bound all-heavy-atom carrier and a complete interaction record. Selection is permitted only through the prospectively registered joint physical order; target structures and comparator outputs remain inaccessible until the applicable artefact has been sealed.

The representation and custody layers are executable. All twenty standard residue codes have current-baseline coordinate constructors. On the 31-residue development control, the corrected carrier contains 239 heavy atoms, 244 bonds, all 28,441 unordered atom pairs and all 6,740,517 shared-atom correlations. The exact fixed-geometry compiler contains 572 electron cells, 126 held electrons, 33 positive sources and all 163,306 distinct electron-pair responses.

Certified search has progressed through registered boundaries of 6, 354, 10,266, 215,940, 3,347,070 and 42,173,082 candidates. The largest completed parent contains 375,014 leaves, 41,798,068 certified prunes and 205 retained frontier states across 11,012 disjoint shards. Independent reconstruction reports zero failures and zero symbolic fallbacks. A separate high-attraction census exactly evaluates 583 occupations, explicitly dominates 573 and retains 10, including eight new nondominated incumbents. These results close their registered parent and high-attraction boundaries; the lower attraction bands, complete fixed-geometry frontier and torsion–protonation–solvent recurrence remain active work.

The empirical development record is deliberately mixed and every classification is method-bound. The corrected 4APD reconstruction, fresh 4B19 baseline and unseen 8HJC transfer are adverse for the historical or partial selectors they tested; none is a primary test of the still-incomplete six-axis joint system. Subsequent work corrected the 4APD carrier and improved the 4B19 median TM-score, but did not erase their original results. The prospectively sealed narrowed 1JDM transfer is positive development evidence, with median official full-chain TM-score 0.45609 across 16 deposited NMR models. A later target-free partial-coordinate recurrence closes on all three current sequences, but improves matched structural TM-score only on 4B19. The official TM-align evaluator and AlphaFold 3 source environment are frozen; restricted comparator parameters and complete matched inference remain unavailable. The publication-integrated workspace suite passes 898 tests with no failures.

The cumulative evidence supports a strong programme forecast: the registered finite exact-computation boundaries are likely to continue closing because the same target-free compiler, partition rules and independent verification have scaled monotonically across six orders of candidate count without an unresolved exact comparison. The programme further predicts that the completed whole-chain recurrence can be subjected to generalised blind parity against the frozen AlphaFold comparator. That parity outcome is a prospective empirical test, not a result of this preliminary version. This Zenodo record will be version-updated when the full investigation, development controls and matched campaign are complete, preserving this preliminary chronology and every favourable, adverse, null and unavailable record.

Detailed preliminary-results synopsis

Protein folding is a decisive test for a theory that claims to derive physical, chemical and biological structure without fitted laws or target-selected rules. This paper reports the present state of the Smithian Fold Theory V3 Protein Fold computational proof programme. The method begins with a registered sequence and condition carrier, constructs a finite 24-sector torsion-word surface, emits a source-bound all-heavy-atom coordinate carrier, retains every atom-pair and shared-atom correlation, and requires a jointly composable, condition-bound descent relation before any representative structure may be selected. Target coordinates and comparator outputs are prohibited until the corresponding V3 artefact has been sealed.

The representation and custody layers are executable. All twenty standard residue codes have current-baseline coordinate constructors. The registered rotor surface contains 38 side-chain variables and 72 outer-carrier pairs; 912 single-rotor sector cases and 25,416 certified display-interval components have been independently checked. Scale, condition, uncertainty and interface custody retains 17 source records, 60 local assignments, 1,600 ordered residue interfaces, 218 protonation states and 47,524 protonation interfaces. On a 31-residue development control, the corrected carrier contains 239 heavy atoms, 244 bonds, all 28,441 unordered atom pairs, 56,882 endpoint incidences and 6,740,517 shared-atom correlations. Independent verification reconstructs the 25,691,185-byte canonical ledger without failure.

The current certified full-V3 result remains a sealed capability halt, not a fold prediction, but the physical joint-state layer has materially advanced. The held condition does not supply an invented direct selector: 218 typed protonation states, all 343 AGA state products and 4,752 one-water states per carrier remain plural until the joint physical order acts. The standard exact fixed-geometry compiler contains 572 electron cells, 126 held electrons, 33 positive sources and all 163,306 distinct electron-pair responses while preserving separate protein and solvent quotas. Response-preserving spin marginalisation and partitioned exact-algebraic frontier solvers agree with exhaustive controls. The full execution is registered as 118 exchange-sector vectors and 350 first-level shards. Two registered real 572-cell deep shards are complete and independently verified. The six-candidate first shard is nested inside the 354-candidate second shard. A disjoint third shard closes 10,266 candidates with 154 leaves, 10,112 certified prunes and 46 retained frontier states. Their verified parallel continuation first closed one complete 215,940-candidate parent prefix. The current fine-grained extension closes one complete 3,347,070-candidate parent prefix across 789 disjoint shards. Its registered successor is now also complete: 11,012 disjoint shards cover 38,826,012 new candidates and close one 42,173,082-candidate parent with 375,014 completed leaves, 41,798,068 certified-pruned candidates and 205 retained frontier states. An implementation-distinct verifier reconstructs every shard, candidate partition, state identity, archive and final frontier with zero failures and zero symbolic fallbacks. This completes the registered parent boundary, not the full fixed-geometry sector or global frontier. A separate target-free whole-sector pass now supplies 22 independently reconstructed exact feasible incumbents from 100 deterministic routes. These incumbents span the full q58/q5 attraction–repulsion response but do not establish frontier completeness. Three registered scalability controls are adverse: a depth-12 census pruned none of 4,096 prefix surfaces, a degree-strengthened depth-8 census pruned none of 256, and a complete 59-stratum consensus-count census pruned no stratum. Each adverse census was independently reexecuted. The records therefore direct the next solver step towards coupled response bounds rather than further repetition of independent attraction and repulsion extrema. The first coupled scalar envelope nevertheless excludes none of the 22 incumbent bands, while certifying the highest-attraction incumbent as the exact linear optimum. A new 1,143-node exact spatial hierarchy validates a quota-preserving dynamic lower bound over all 572 cells. Its standalone repulsion floor (97.42645...) is weaker than the prior degree floor (147.61975...) and remains below the lowest incumbent response (378.84495...); the overbroad original “improves bound” label is preserved and corrected in a reconciliation record. The coupled hierarchy dynamic programme was then executed and independently reexecuted, but excluded none of the 22 open response bands. A stronger complete-sector cardinality-RLT relaxation, with all three McCormick inequalities, then raises the exact certified repulsion floor to 192.197997235382... This is strictly stronger than the prior 147.619752821... floor but remains below the least feasible incumbent 378.844950488... Its separately frozen verifier independently reconstructs the rational dual with zero failures. It is therefore a positive bound result, not minimum or frontier closure. An exact attraction-conditioned RLT test of the highest band then returns 1688.211059043998..., below its 1706.611377155384... exclusion ceiling. Independent verification passes with zero failures. The optimum is the preceding feasible incumbent, so the adverse result isolates the exclusive attraction boundary rather than a repulsion defect. A first exact symbolic top-rank implementation was then preserved after a controlled 4,894-second performance interruption without an outcome. Its interval-first V2 completed in 49.7 seconds and proved that the seeded rank-20 incumbent is not the actual runner-up. The resulting complete threshold census contains 583 occupations: 582 definitely above the boundary and one boundary-overlap state. Exact response evaluation explicitly dominates 573 and retains 10. Two retained endpoints were already seeded; eight are new nondominated incumbents. Independent reexecution reconstructs all 583 repulsions with zero failures, closing this high-attraction slice while leaving the lower bands open. Complete fixed-geometry coverage and torsion recurrence remain open. A separate sequence-only development track produced three prospectively sealed short-protein records. The fresh V2-generalised baseline on 4B19 was adverse. A post-target alpha-helical diagnosis then transferred to two unseen targets: it was adverse on the six-cysteine 8HJC holdout and positive on the narrowed zero-cysteine 1JDM holdout, whose median official full-chain TM-score over all 16 deposited NMR models was 0.45609. These are development results, not the certified V3 solver or the primary parity campaign. A later target-free partial-coordinate recurrence closes independently on all three now-open sequences, but its structural TM-score improves on only 4B19 and regresses on 8HJC and 1JDM.

The corrected historical 4APD reconstruction control remains adverse: $tm_repo = 0.08579343893330212$, IDDT-C-alpha 0.5241935483870968, C-alpha RMSD95 6.25666027980953 A, Kabsch C-alpha RMSD 6.515371977465647 A and C-alpha dRMSD 5.18633263505115 A. The official TM-align evaluator and the pinned AlphaFold 3 source environment are now frozen. AlphaFold's official data tests pass 7/7 locally and the experimental Apple MPS device is visible; the upstream inference suite records one pass and 16 errors because its example tests request a CUDA-labelled backend and the restricted model parameters are absent. The latest complete recorded workspace suite has 898 passing tests and no failures. These checks establish implementation, custody and development chronology; AlphaFold parity remains unresolved.

Headline findings

Result	Current evidence	Scientific status
Standard residue constructor surface	20 of 20 residue codes	Implementation complete at the current source-bound baseline
Ordered side-chain rotor surface	38 variables; 72 outer-carrier pairs	Exact source-graph census complete
Sector renderer	912 single-rotor cases; 25,416 interval components	Primary and independent checks pass
Condition and interface custody	17 sources; 60 local assignments; 1,600 interfaces	Complete at the registered plural-custody boundary
Protonation and transfer custody	218 states; 47,524 interfaces; 114 strict local transfer edges	Complete plural surface and exact named-proton transfer graph; condition order remains open
Corrected 4APD atomistic carrier	239 atoms; 244 bonds	Development reconstruction control, not a V3 prediction
Complete pair surface	28,441 pairs; 6,740,517 shared-atom correlations	No cut-off; independently verified
Carbonyl correction	7 sub-0.5 A pairs before; 0 sub-1 A pairs after	Clerical geometry defect corrected; predecessor preserved
Generic transition probe	84 states; 6,972 directed transitions; 1 recurrent class	Adverse non-discrimination result
Exact partial physical coordinates	Nuclear repulsion and source-formal-charge response	Executable, torsion-dependent partial coordinates; electrons, screening and complete atom-state charges remain open
Componentwise partial order	24 states; 240 dominance edges; 3 nondominated states	Exact no-fitted-weight result; no unique representative
Partial recurrence	552 ordered pairs; 240 descent edges; closed classes 21, 22 and 23	Exact plural terminal ensemble at the partial-coordinate boundary
Exact-point exclusion	24 of 24 states admissible	Hard equality boundary complete; no finite exclusion radius or exchange height claimed
State-dependent-height route census	17 routes; 17 decisions; 0 complete-height survivors	Current repository authority exhausted without type promotion
Historical physical-coordinate completion census	6 structural laws; 0 of 6 then-complete executable protein coordinates	Preserved predecessor to the authorised joint-response extension
Current six-axis joint relation	256 forms; 1 preserving survivor	All six preserving axes constructed at the declared fixed-geometry boundary; full execution remains open
Exact plural joint carrier	343 protein carriers; 4,752 water states per carrier	Complete at the declared AGA boundary
Standard fixed-geometry compiler	572 cells; 126 electrons; 33 positive sources; 163,306 pair responses	Exact and independently verified
Full execution plan	118 sector vectors; 350 first-level shards	Exact disjoint candidate partition complete; execution open
First registered real deep shard	6 candidates; 2 leaves; 4 certified-pruned; 1 frontier state	Complete and independently verified; not a complete sector or fold
Second registered real deep shard	354 candidates; 32 leaves; 322 certified-pruned; 13 frontier states	Complete and independently verified; three implementation-performance aborts preserved as non-empirical records
Third registered real deep shard	10,266 candidates; 154 leaves; 10,112 certified-pruned; 46 frontier states	Complete, disjoint from the second shard and independently verified
Previous exact parent prefix	3,347,070 candidates; 40,899 leaves; 3,306,171 certified prunes; 135 frontier states	Exact independently verified 789-shard predecessor
Current exact parent prefix	42,173,082 candidates; 375,014 leaves; 41,798,068 certified prunes; 205 frontier states	Exact independently verified 11,012-shard reconstruction; not a complete sector or fold
Current full-solver gate	capability_halt	Six-axis relation and generic solver constructed; remaining fixed-geometry frontier and torsion recurrence open; zero target accesses
Corrected 4APD comparison	TM-like diagnostic 0.08579343893330212	Adverse historical reconstruction control
Fresh 4B19 baseline	Official TM-score median 0.17500 over 5 NMR models	Adverse sealed development baseline

Result	Current evidence	Scientific status
First unseen sequence-only holdout, 8HJC	Official TM-score median 0.18242 over 20 NMR models	Adverse blind development evidence
Narrowed unseen holdout, 1JDM	Official TM-score median 0.45609 over 16 NMR models	Positive blind development evidence for the registered narrowed class
Generalised partial-coordinate panel	Recurrence closure 3/3; structural TM-score improvement 1/3	Computational generalisation established; structural-accuracy generalisation adverse
Canonical full-chain evaluator	Official TM-align source and executable hash-bound	External TM-score identity complete
Local AlphaFold 3 environment	Data tests 7/7; MPS visible; inference tests 1/17	Source environment verified; restricted weights and full databases absent
Current verification	898 tests passed; 0 failed	Computational and publication-audit verification only
AlphaFold generalised blind parity	No campaign result	Not established

Current status

Formally implemented. The finite representation, plural protonation and explicit-solvent carrier, source-bound coordinate emission, exact fixed- geometry response compiler, partition-preserving algebraic solver controls, resumable shard plan, complete atom-pair ledger, local metric kernel, strict 100-row parity decision and capability-halt branch are implemented. The real deep-shard receipts and fine-grained continuation reconstruct one exact 42,173,082-candidate parent prefix independently across 11,012 shards. The full fixed-geometry sector and global frontier are not complete.

Current empirical status. The already-open historical 4APD reconstruction remains adverse. Three short-protein development predictions were sealed before target access: the 4B19 baseline was adverse, the first unseen sequence-only holdout was adverse, and the narrowed second unseen holdout was positive within its registered cysteine-bounded hydrophobic-majority class. None is a primary campaign row or a certified full-V3 predictor result. The later generalised partial recurrence closes on all three sequences, but its matched structural accuracy improves on only one of three.

Chronology. The parity decision was frozen before primary target selection. The interaction ledger and capability result were sealed without target access. The 4APD target belongs to the preserved historical development partition and cannot become a blind holdout. The 4B19 baseline, 8HJC holdout and 1JDM holdout each retain independent pre-target seals; later diagnostic reasoning is not backdated into an earlier prediction.

Current scope. This is a frontier validation application of admitted SFT claims. It is not a new model-admitted Biology theorem and does not alter the frozen SFT engine or verification authority.

Publication status. This is a publication-ready preliminary-results paper, not the final parity paper. No certified general protein-fold solver, primary 100-target campaign result, AlphaFold match, AlphaFold parity result or field-wide empirical confirmation is claimed.

Authorising records. The current state is bound by `workspace_manifest.json`, `audits/continuation_checkpoint.json`, `audits/whole_chain_interaction_custody_v1.json`, `audits/current_scientific_gate_v20.json`, `audits/seeded_parallel_parent_extension_v3.json`, its implementation-distinct verification, and the primary and independent `full_sector_high_band_response_census_v1` records. The two registered incomplete operations are controlled by `spec/full_sector_attraction_subband_bisection_v2.json` and `spec/seeded_parent_high_band_reconciliation_v1.json`; their incomplete outputs are excluded from this publication snapshot and no result is imported from an unfinished temporary stream.

1. Public scientific mission and authorship boundary

Ernos Labs is an open scientific movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is to make scientific authority narrow, inspectable and revocable. A result should be open to replication and criticism through its dependencies, generated alternatives, eliminations, controls, source custody, executable traces and retained adverse evidence. Credentials may help route review; they do not replace a derivation or rescue a failed control.

Maria Smith developed Smithian Fold Theory outside the conventional academic and institutional route. That history is not evidence for any scientific statement. It is part of the project's institutional argument: a system that confuses permission, prestige or funding with evidence excludes possible questions and contributors before their work can be tested. SFT therefore separates open criticism from scientific admission and makes both publicly inspectable.

Maria Smith retains scientific authorship, ownership and publication authority. Code and editorial assistance produced by automated agents do not confer scientific authorship, authority to admit a claim or authority to publish. Maria Smith authorised this updated preliminary Zenodo version and the matching push to the repository's main branch, and directed that later complete results be issued as further versions of the same record. That action-specific authorisation does not extend to a final parity claim or any unrelated remote record.

2. Scope and ownership boundary

The primary application carrier is one canonical, unmodified, single-chain protein sequence of 20–800 standard amino-acid residues under one declared condition record. The intended output is the complete recurrent structural support at a declared finite resolution, together with its transition, elimination, source and resource records. A single representative is permitted only if a target-independent rule has already been derived and sealed.

Biology owns the sequence-to-fold and fold-to-ensemble relations through:

- SFT-BIO-PROTEIN-SEQUENCE-001;
- SFT-BIO-PROTEIN-FOLD-001; and
- SFT-BIO-PROTEIN-ENSEMBLE-001.

The application extension is registered as SFT-APP-PROTEIN-INTERACTION-DESCENT-001. It is an authorised frontier application registration, not an engine-admitted scientific claim. Chemistry and Physics retain ownership of molecular identity, geometry, interaction, charge, exclusion, state order, binding, solvation, proton transfer, dipole and polarisation. Mathematics and computation retain ownership of exact finite carriers, partial orders, paths, recurrence, frontier custody and predecessor records.

The application excludes complexes, covalent modifications, membrane insertion, chaperone assistance, trafficking, protein–nucleic-acid recognition and cellular rescue. These remain separately named Biology frontiers. Their absence from this campaign does not classify them as scientifically absent.

3. Terminology and evidence constitution

In this paper, a **theorem** is an exact formally admitted result and a **law** is an admitted dependency relation. A **claim** is a registered scientific statement with its own authority and evidence status. A **constitution** is the declared carrier, relation, evidence and decision boundary under which a result is interpreted. A **derivation** is a generated consequence of admitted carriers and relations. A **prediction** is a prospectively sealed consequence whose target was inaccessible before sealing. An **observation** is a source-bound external record; a **measurement** remains bound to its instrument, method, condition and source. A **reconstruction** uses an already-known or historically available target partition.

A **correspondence** is a relationship examined only after derivation. An **exact numerical correspondence** equates exact values, a **structural correspondence** relates forms without numerical identity, and a **boundary correspondence** relates only the declared limiting or interface surface. **Support** is relevant but non-unique evidence. **Confirmation** is reserved for an evidence protocol whose registered decision rule warrants that status; no result in the present protein campaign has that status.

Verification means that an implementation, archive or receipt satisfies its registered mechanical checks. **Validation** is reserved for the applicable empirical protocol. **Compatibility** is non-adverse but non-discriminating evidence. An **adverse result** conflicts with the registered test or fails to discriminate when discrimination is mandatory. An **unresolved result** lacks required evidence. A **capability halt** is the registered output when required coordinates or resources remain unavailable; it is neither a prediction nor an empty-support result.

An **implementation identity** is the exact version and hash surface of an executable object. **Formal status** records derivational admission, **computational status** records implementation and verification, **empirical status** records evidence under its chronology, and **publication status** records editorial approval and release authority. None substitutes for another.

Formal closure names a completed exact grammar. **Implementation closure** names executable and verified code.

Current-evidence closure is always bounded to a frozen evidence surface. **Foundational closure** closes the named dependency foundation, while **field-wide closure** closes every registered family within the named field boundary; neither is inferred here from an application test. **Extension openness** means that future applications, measurements, corrections and lawful invalidations remain open. Every use of *complete* below names the completed object; it never means the end of protein science.

4. Mathematical constitution

4.1 Sequence and condition carrier

Let a sequence be the ordered word

Sequence carrier: $S = (a_1, a_2, \dots, a_n)$, with a_i in A .

where A contains the twenty registered standard amino-acid identities. Each position retains chirality, modification, ambiguity and source fields. The primary campaign admits explicit L chirality for non-glycine residues and an achiral record for glycine. A condition carrier retains temperature, pressure, pH, solvent, ligand, method and source fields; unavailable fields remain unavailable rather than receiving defaults.

4.2 Torsion-word surface

Every registered backbone or side-chain rotor is assigned one of twenty-four ordered sectors. If $r(S)$ is the complete side-chain rotor count, the word width is

Word width: $k(S) = 3(n - 1) + r(S)$.

and the raw finite word count is

Raw word count: $N_{\text{word}}(S) = 24^{k(S)}$.

The product is represented lazily; its size is not confused with an executed exhaustion. Exact sector reversal and source-graph automorphism actions are retained. No word is selected by a target score, hash order or lexicographic identity.

4.3 Source-bound coordinate emission

Backbone coordinates use the registered CCP4/PROCHECK residue-class means for N-CA, CA-C, C-N, N-CA-C, CA-C-N and C-N-CA. Side-chain heavy atoms retain the wwPDB Chemical Component Dictionary ideal-reference shape in the local N-CA-C frame. The source values are conventional coordinate records; they are not exact SFT proof values.

The first implementation left a carbonyl oxygen in its local frame while the successor nitrogen moved with the peptide torsion. The resulting seven sub-0.5 Å atom pairs were detected by the complete interaction ledger. The corrected rule retains the same source C-O length and CA-C-O angle and places each nonterminal oxygen trans to the successor nitrogen in the peptide plane. The defective carrier and its archive remain preserved.

4.4 Complete atom-pair custody

For m retained heavy atoms, every unordered pair is emitted:

Unordered-pair count: $N_{\text{pair}} = C(m, 2) = m(m - 1)/2$.

Every pair retains atom identity, residue position, elements, source formal charge inscriptions, shortest polymer-graph separation, residue separation and the source-bound squared distance as a canonical floating-point hexadecimal record. No contact or distance cut-off is applied.

The complete shared-atom correlation count is

Shared-atom correlation count: $N_{\text{corr}} = m C(m - 1, 2) = m(m - 1)(m - 2)/2$.

For the corrected 4APD control, $m=239$, so $N_{\text{pair}}=28,441$ and $N_{\text{corr}}=6,740,517$. Endpoint incidence is exactly $2N_{\text{pair}}=56,882$.

4.5 Registered descent form

The authorised application extension generated the complete product of eight binary preservation axes. All 256 forms were decided; 255 were rejected and one relation form survived. The survivor retains the complete sequence–condition–atomistic conformation carrier, holds solvent, protonation, temperature and method, compares a complete physical and chemical relation vector componentwise, requires at least one strict improvement, preserves incomparability, retains complete all-body correlation, emits every nondominated recurrent class, and adds no fitted trade-off.

This is closure of the **relation form**, not closure of its protein-specific coordinates. A grammar can uniquely force the form of a comparison while the values required to execute that comparison remain unavailable.

4.6 Transition and recurrence probe

The least committed transition replaces exactly one torsion-word coordinate with every other registered value. For local domains L_i , bridge domain B and length n , the successor count is

$$\text{Successor count: } \text{sum}_i(|L_i| - 1) + (n - 1)(|B| - 1).$$

The complete one-residue glycine boundary contains 84 words and $84 \times 83 = 6,972$ directed transitions. Exhaustion yields one closed recurrent class containing all 84 words. This is an adverse non-discrimination result: reachability, reversibility and predecessor custody do not select a folded class.

5. Dependency structure

The direct application dependency inventory is:

Domain	Direct claim identities	Role in this application
Biology	SFT-BIO-PROTEIN-SEQUENCE-001; SFT-BIO-PROTEIN-FOLD-001; SFT-BIO-PROTEIN-ENSEMBLE-001	Sequence, condition-bounded fold and observed ensemble carriers
Application	SFT-APP-PROTEIN-INTERACTION-DESCENT-001	Registered frontier relation form; not model-admitted
Molecular constitution	SFT-CHEM-MOL-MOLECULE-001; SFT-CHEM-MOL-GEOMETRY-001; SFT-CHEM-MOL-INTERMOLECULAR-001	Complete molecular carriers, geometry and interaction channels
Electronic structure	SFT-CHEM-MOLECULAR-EXCLUSION-EXCHANGE-006; SFT-CHEM-STATE-ENERGY-ORDER-004; SFT-CHEM-MOLECULAR-DIPOLE-MAGNITUDE-005; SFT-CHEM-MOLECULAR-POLARIZABILITY-006	Exchange, finite state order, dipole and response relations
Configuration and binding	SFT-CHEM-MOLECULAR-STATE-TRANSITION-009; SFT-CHEM-CONFIGURATION-ORDER-PATH-011; SFT-CHEM-INTERMOLECULAR-BINDING-011	State transitions, paths and bound/separated order
Solvation and direction	SFT-CHEM-SOLVATION-DISSOLUTION-FREE-ORDER-015; SFT-CHEM-FREE-ENERGY-EQUIVALENT-DIRECTION-007	Condition-bound transfer and componentwise direction
Acid–base	SFT-CHEM-AB-ACID-BASE-001; SFT-CHEM-AB-PROTON-TRANSFER-001; SFT-CHEM-AB-LEWIS-001	Protonation identity and transfer relations
Physics	SFT-PHYS-FIELD-COULOMB-GAUSS-CLOSURE-003; SFT-PHYS-THERMO-EQUILIBRIUM-001	Held-charge inverse-square response and equilibrium boundary

Each claim retains its immutable registration and model-admitted receipt where applicable. The frontier application registration does not modify those claims or the frozen admission engine.

6. Registered source surface

The atomistic source surface contains twenty current standard-residue wwPDB CCD captures and one parent-bound amino-acid variant dictionary. All 21 registered transports succeeded. The standard component census retains 387 atoms, 373 bonds, 60 leaving-atom rows, 21 stereochemical-atom rows and 26 aromatic-atom rows. The heavy-atom baseline used by the interaction census contains 167 non-leaving identities: 164 zero formal-charge inscriptions and three positive inscriptions (H:ND1, K:NZ, R:NH2). There is no opposed baseline inscription and no complete partial-charge field.

The main-chain geometry capture retains published CCP4/PROCHECK class means and uncertainties. The source-bound coordinate layer distinguishes source values from exact derivational equality. Failed transports and missing uncertainty fields remain explicit. Twenty side-chain uncertainty records are unavailable; none was imputed.

The AlphaFold comparator registration freezes the official AlphaFold 3 source commit `85c4d20505fd5cef05eac22b534d4e793971ae69`, the weights URI and GCS generation, historical database versions, five deterministic seeds, five samples per seed and the official highest-ranking representative. A deterministic source archive is preserved with SHA-256 `f81dd985fe4ad0d24a365c47f5740bde904eb95bd7ab6c2ae5a366dcc58df93a`. The pinned Python environment is installed, the official data suite passes 7/7, HMMER is present and the experimental Apple MPS device is visible. The model-parameter SHA-256, registered full database installation and complete comparator inference remain unavailable.

7. Derivational and implementation sequence

7.1 Exact local geometry and polymer graph

The kernel begins with canonical sequence identity, exact rational direction grammar, local N–CA–C frames, explicit chirality and peptide adjacency. The standard peptide graph removes the registered terminal leaving atoms, retains the released water carrier and adds every ordered peptide C–N bond. Exact local support, peptide boundaries and nonlocal residue-pair records are evaluated on the same whole-chain candidate rather than as independent one-residue witnesses.

7.2 Side-chain closure programme

Successive registered grammars close alanine and glycine, cysteine and serine, valine, threonine, proline, isoleucine, leucine, aspartate and asparagine, glutamate and glutamine, methionine, lysine, arginine, phenylalanine, tyrosine, histidine and tryptophan at their declared boundaries. The programme preserves equal-unit adverse results, mirror-closure failures, storage and preflight aborts, corrected registrations and implementation-distinct checks. The final uniform current-baseline catalogue covers all twenty residue codes; it does not claim that one ideal reference is a universal physical conformer.

7.3 Rotor, word and sector mapping

The ordered rotor census partitions every retained source bond and supplies 38 side-chain variables. Polymer templates append `phi`, `psi` and `omega`. The lazy 24-sector word surface preserves complete variable order, exact reversal and source-graph symmetry. The sector renderer preserves every source-graph bond distance, rotation-axis endpoint and fixed atom over 912 single-rotor cases. Primary and independent reconstructions agree.

7.4 Scale, condition and interface custody

Seventeen geometry and condition records map to 60 local assignments and 1,600 ordered residue interfaces. The complete plural protonation surface contains 218 states and 47,524 ordered interfaces. No fitted pH rule chooses one state. The exact named-proton graph adds 114 strict local edges across 163 states and preserves 55 isolated states; all 12,996 donor-by-acceptor templates remain in the audit archive. This closes occurrence and lawful local-transfer custody but deliberately leaves condition-specific selection and hydrogen geometry open.

7.5 Whole-chain carrier and correction

The word renderer composes the registered backbone means, peptide interfaces, local CCD reference shapes and torsion sectors. Its first full development execution generated 239 atoms and 244 bonds before the historical target was opened. Complete pair enumeration then identified the carbonyl-plane error. The source-preserving corrected constructor removes all sub-1 Å pairs on the same control while leaving every C-alpha coordinate and metric unchanged.

7.6 Interaction ledger and execution gate

The interaction ledger emits all atom pairs and both endpoint incidences, classifying graph separation without assigning a physical interaction class. Formal charges are not promoted to partial charges. The historical partial gate found two torsion-dependent coordinates, 240 dominance edges and three nondominated recurrent classes, while exact-point exclusion accepted all 24 states and the then-current route census found no complete height.

The later authorised joint-microstate derivation supersedes that historical 0/6 result as the current constructor inventory. Exhaustive decision of 256 candidate forms retains one preserving relation with separate axes for held condition and proton ownership, elementary charge response, exclusion and exchange, polarisation through explicit electron redistribution, explicit solvent, and componentwise order. The standard fixed-geometry compiler and generic certified frontier solver are complete and independently verified. The current `capability_halt` is therefore an execution boundary: the complete fixed-geometry frontier and the torsion–protonation–solvent recurrence remain open.

7.7 Evaluator and parity decision

The local evaluator computes IDDT-C-alpha, C-alpha RMSD95, Kabsch C-alpha RMSD, dRMSD and the explicitly labelled historical `tm_repo` diagnostic. `tm_repo` performs one full-chain Kabsch fit and is not substituted for the registered external full-chain TM-score. Official TM-align version 20220412 is now source-captured, compiled and hash-bound. Its campaign normalisation is the full registered sequence length supplied with `-u`; every deposited NMR model is retained. The 100-row decision kernel requires twenty targets in each of five length strata and retains complete, failed, invalid and unavailable rows.

Parity is a strict conjunction: V3 representative coverage must be 100/100; median full-chain TM-score and IDDT-C-alpha must meet or exceed AlphaFold 3; median C-alpha RMSD95 must be no worse; V3 must win or tie on TM-score and IDDT-C-alpha on at least 50 targets; median TM-score must meet or exceed the comparator in every length stratum; and no row may be censored or replaced.

8. Evidence chronology and controls

Date / order	Record	Evidence class	Result
Historical V2	Published 24-structure panel	Mostly ubiquitin-family reconstruction; target-derived material relation	Median <code>tm_repo</code> 0.9255486262; not generalised evidence
Historical V2	Fresh diverse four-target panel	Development-observed generalisation control	Adverse: mean TM 0.0742934939, median 0.0781022432, maximum 0.0987934258; median RMSD95 14.117 Å
29 July 2026	V3 parity rule	Preregistered before target selection	Decision rule frozen; no target set selected
29 July 2026	No-extra-rule recurrence probe	Exact exhaustive control	Adverse: one recurrent class retains all 84 glycine words
30 July 2026	First 4APD V3 carrier	Historical reconstruction development control	Seven sub-0.5 Å carbonyl-related pairs detected
30 July 2026	Corrected 4APD V3 carrier	Corrected development reconstruction	Zero sub-1 Å pairs; complete archive verified
30 July 2026	Corrected 4APD target evaluation	Post-seal historical development comparison	Adverse TM-like and RMSD result; C-alpha metrics unchanged by correction
30 July 2026	Predictor gate execution	Development capability result	Sealed <code>capability_halt</code> ; zero target accesses; no prediction artefact
30 July 2026	Authority census	Current repository assessment	Six structural laws present; zero executable protein coordinate families
30 July 2026	Exact partial-coordinate programme	Target-free formal and computational evidence	Two torsion-dependent partial coordinates; 240 dominance edges; three nondominated states
30 July 2026	Named-proton transfer graph	Exact exhaustive implementation evidence	114 strict local edges; 55 isolated states preserved
30 July 2026	Conformation-order obstruction	Exact exhaustive theorem and general proof	A unique selector requires a state-dependent height or equivalent relation
30 July 2026	Renewed completion census	Current repository assessment	Partial advances preserved; zero of six complete coordinate families
30 July 2026	Authorised joint-microstate relation	Target-free application derivation	256 forms decided; one preserving six-axis relation survives
30 July 2026	Fixed-geometry compiler and solver	Exact implementation evidence	Six preserving axes executable at the declared standard AGA boundary; generic solver verified

Date / order	Record	Evidence class	Result
30 July 2026	Resumable parallel parent execution	Exact implementation evidence	One 215,940-candidate parent prefix complete with 107 retained frontier states
30 July 2026	Fine-grained seeded parent extension	Exact implementation evidence	One 3,347,070-candidate parent prefix complete across 789 shards with 135 retained frontier states; independently reconstructed
31 July 2026	Registered 42,173,082-candidate parent extension	Exact implementation evidence	All 11,012 shards close; 375,014 leaves, 41,798,068 certified prunes and 205 retained frontier states; independently reconstructed with zero failures and zero symbolic fallbacks
30 July 2026	Exact partial recurrence	Independent current-state reconstruction	Three all-pair closed classes; four cyclic-neighbour closed classes
30 July 2026	Height-route census	Hash-bound current-authority trace	17 routes decided; zero complete-height survivors
30 July 2026	Predictor gate v2	Sealed current development capability result	Independently verified capability_halt; zero target accesses
30 July 2026	Fresh 4B19 V2-generalised baseline	Prospectively sealed development baseline	Adverse: official median TM-score 0.17500 over 5 NMR models
30 July 2026	4B19 alpha-helical diagnosis	Post-target development analysis	Positive diagnostic; lawfully motivates a new unseen test only
30 July 2026	Unseen 8HJC holdout	Sealed blind development evidence	Adverse: official median TM-score 0.18242 over 20 NMR models
30 July 2026	Narrowed unseen 1JDM holdout	Sealed blind development evidence	Positive within registered class: official median TM-score 0.45609 over 16 NMR models
30 July 2026	Generalised partial-coordinate three-target panel	Post-target development evidence	Recurrence closes 3/3; structural median TM-score improves 1/3 and regresses 2/3
30 July 2026	Canonical TM-align freeze	Pre-primary-campaign evaluator registration	Source, portability change, executable and parser identities closed
30 July 2026	AlphaFold 3 local environment audit	Comparator readiness evidence	Data tests 7/7; MPS visible; inference suite 1/17; restricted assets absent
Pending	V3 100-target blind panel	Sealed unseen-sequence holdout	Not started
Pending	Matched AlphaFold 3 panel	Frozen comparator	Not executed

9. Corrections, adverse results and unavailable evidence

No adverse label in this paper is a free-standing verdict on the complete V3 programme. Each label is bound to the exact carrier, selector, control and chronology that produced it. Later work can supersede an implementation or constructor without rewriting the observed predecessor result. Conversely, stronger exact computation cannot be used as an empirical rescue before the completed system produces its own sealed prediction.

Earlier or adverse boundary	Stronger subsequent work	Current standing
V2 fresh four-target generalisation was adverse	V3 replaced the target-derived material selector with target-free representation, complete interaction custody and the six-axis joint relation	The V2 outcome remains historical adverse evidence; it is not a test of the unfinished full-V3 predictor
Generic 84-state transition probe retained one undifferentiated recurrent class	The later 256-form derivation retains one jointly composable six-axis physical relation	The generic transition is superseded as the active selection route, while its adverse non-discrimination result remains valid
First 4APD carrier contained seven sub-0.5 Å pairs	The corrected carbonyl-plane constructor removes every sub-1 Å pair without changing the registered source geometry	The defective constructor is superseded; its detection and archive remain historical evidence
Corrected 4APD reconstruction scored adversely	The exact compiler, certified frontier search and current joint relation replace that reconstruction route	The structural score remains adverse for 4APD; no full-V3 prediction yet exists to supersede it empirically

Earlier or adverse boundary	Stronger subsequent work	Current standing
Historical current-authority census found 0/6 complete physical coordinate families	The authorised joint-microstate derivation constructs all six preserving axes at the declared fixed-geometry boundary	Superseded as the current constructor inventory; preserved as the historical census
Independent-extrema, degree-strengthened and 59-stratum pruning controls excluded nothing	Coupled bounds, a complete cardinality-RLT certificate and exact high-band evaluation were implemented	The failed pruning strategies are superseded operationally; their adverse outcomes remain reproducible controls
The first spatial-hierarchy floor was weaker than the previous degree floor	The cardinality-RLT floor increased the certified repulsion bound from 147.619752821... to 192.197997235382...	The overbroad "improves bound" wording is corrected; the weaker exact certificate remains preserved
Coupled envelopes and the attraction-conditioned RLT test did not close the highest band	Exact evaluation of 583 threshold occupations dominates 573 and retains 10, closing the registered high-attraction slice	Those bounds remain adverse for exclusion; their scientific obstacle is superseded at the closed high-attraction boundary, not in the open lower bands
Fresh 4B19 and unseen 8HJC sequence-only transfers were adverse	The target-free partial recurrence closes on both sequences and improves 4B19 from 0.17500 to 0.23779	4B19 is materially improved but remains development evidence; 8HJC regresses to 0.139675, so its adverse status remains active
Narrowed unseen 1JDM transfer was positive at 0.45609	The later generalised partial recurrence closes but scores 0.245185	The earlier sealed positive result remains valid for its registered narrow class; the later selector regression is separately adverse

The following records therefore remain scientifically active and visible:

1. The favourable V2 24-structure result is historically real within its registered reconstruction surface, but the selector used target-derived material information and the panel was not generalised.
2. The V2 fresh diverse four-target panel is adverse. It is not removed by the later V3 implementation.
3. Exhaustive reversible one-coordinate transitions do not discriminate a folded recurrent class.
4. The first V3 carbonyl mapping was defective. Its ledger, seven impossible short pairs and path-normalisation abort are preserved. The corrected version supersedes it for current coordinate emission.
5. The corrected 4APD C-alpha result remains adverse. A geometry correction is not reclassified as a successful fold.
6. The historical coordinate census found all six executable families unavailable. That record remains part of the chronology, but the authorised joint-microstate derivation later constructed all six preserving relation axes at the declared fixed-geometry boundary. Full frontier execution and arbitrary-sequence torsion-protonation-solvent recurrence remain unavailable.
7. Official AlphaFold 3 weights, databases and complete execution remain unavailable in the current environment. The source runtime, HMMER, official data tests and experimental Apple MPS backend are verified; the restricted parameters and registered full databases are absent. This is unavailable comparator evidence, not an adverse comparator result.
8. The 4B19 post-target alpha-helical diagnosis is not a blind validation. Its first unseen transfer to six-cysteine 8HJC was adverse and remains so.
9. The narrowed cysteine-bounded rule transferred positively to unseen 1JDM. That result is positive blind development evidence for the narrow registered class and is not promoted to general protein-fold completion.
10. The later target-free partial recurrence closes computationally on 4B19, 8HJC and 1JDM, but improves median structural TM-score only on 4B19. The two regressions remain adverse evidence against its current structural selector.

10. Six-axis physical relation result

The historical census distinguished structural laws from the then-missing values needed to execute them and returned 0/6. That result is preserved at its chronology boundary. The later authorised derivation registered 256 joint relation forms and retained one preserving survivor:

Required physical account	Current preserving construction	Remaining execution boundary
Condition and protonation	Held condition and complete plural proton ownership	Complete arbitrary-sequence joint recurrence

Required physical account	Current preserving construction	Remaining execution boundary
Charge response	Elementary positive and electron occurrences over exact separations	Complete full frontier at every emitted geometry
Exclusion and exchange	Exact occupancy, complementary spin and support-overlap relations	Complete sector frontier and recurrence
Polarisation and dispersion	Explicit electron redistribution and correlated all-body transition support	Complete occupancy frontier at every emitted geometry
Solvent transfer	Explicit finite solvent carriers and complete transfer custody	All torsions, placements and registered boundary extensions
Electronic-state order	Componentwise nonworsening with strict improvement and incomparability	Complete fixed-geometry frontier followed by whole-chain recurrence

Thus all six preserving axes are constructed at the declared standard fixed-geometry boundary without fitted scalar weights, imported partial charges, pair potentials or continuum solvent. The generic exact solver is also complete. The current halt concerns the unexecuted full frontier and torsion–protonation–solvent recurrence; it is not a current 0/6 constructor claim. The machine reconciliation is `audits/full_coordinate_completion_reconciliation_v4.json`.

11. Current claim inventory

Application and Biology claims

Claim ID	Formal status in this programme	Empirical status	Current closure
SFT-BIO-PROTEIN-SEQUENCE-001	Admitted dependency	Retained through source custody	Dependency available
SFT-BIO-PROTEIN-FOLD-001	Admitted dependency	Registered experiment remains open	Foundational law available; application execution open
SFT-BIO-PROTEIN-ENSEMBLE-001	Admitted dependency	No new V3 ensemble observation	Dependency available
SFT-APP-PROTEIN-INTERACTION-DESCENT-001	Frontier application registration; not model-admitted	Development controls only	Relation form closed; coordinate execution open

Direct Chemistry and Physics dependencies

The complete direct IDs are listed in Section 5. Their structural-law status is preserved from their current claim packages. This application neither revises their empirical classifications nor awards a new admission. The six claims used by the live coordinate census are individually hash-bound in `audits/condition_bound_physical_coordinate_census_v1.json`.

Complete claim-level audit

The scientific audit companion paper `/COMPLETE_CLAIM_AUDIT.md` exposes the full publication-guidance fields for all 21 records named by this application: one frontier application registration and 20 model-admitted dependencies. Its manifest is `paper/COMPLETE_CLAIM_AUDIT_MANIFEST.json`. The audit preserves each dependency's own formal and empirical status; inclusion in the application does not transfer admission or promote development evidence.

Version 0.9.4 also reconciles a clerical identity defect in the immutable v0.9.3 prose: two earlier workspace-manifest hashes referred to predecessor states. The corrected current identity is recorded consistently in this paper, the scientific audit and the machine archive. The predecessor values and the non-scientific nature of the correction remain explicit in `paper/PRELIMINARY_V0_9_4_IDENTITY_RECONCILIATION.json`.

12. Reproducibility

Reproduction begins by verifying the frozen engine and verification-authority seals from the repository root:

```
python3 tools/verify_engine_seal.py
python3 tools/verify_verification_authority_seal.py
```

The application test suite is then executed without importing target data:

```
python3 -m unittest discover \
-s applications/frontier/v3_computational_proofs/protein_folding/tests \
-p 'test_*.py'
```

The current result is 898 passed. Detailed commands for source custody, coordinate rendering, interaction archive verification, gate replay and evaluator checks are retained in `paper/REPRODUCIBILITY.md`. Full machine identities are listed in `paper/MACHINE_ARCHIVE_MANIFEST.json`.

Passing tests verify the registered implementations and artefacts. They do not constitute a blind empirical result.

13. Ongoing programme and evidence-based forecast

This preliminary release freezes the evidence available on 31 July 2026 while the computational proof programme continues. Two target-free exact operations remain registered and incomplete at this publication boundary: the midpoint census of the remaining attraction subbands and the seeded-parent high-band reconciliation. Their final counts are not imported into this paper before completion and independent verification.

The programme's forecast separates three outcomes that require different kinds of evidence:

Forecast	Present basis	Required decision evidence
Complete the registered exact fixed-geometry investigation	One compiler and partition grammar has closed successively larger registered boundaries from 6 to 42,173,082 candidates; the largest run has zero independent-verification failures and zero symbolic fallbacks; the high-attraction slice is exactly closed	Complete remaining bands, global frontier reconciliation and implementation-distinct verification
Generalise the exact carrier and recurrence across eligible sequences	The arbitrary-sequence compiler reconstructs the AGA witness and compiles a non-AGA glycine control with 285 cells, 40 electrons and 40,470 pair responses; the partial recurrence closes on 3/3 current development sequences	Complete torsion–protonation–solvent recurrence and pass the frozen development and adverse controls
Achieve generalised blind parity with AlphaFold	Exact representation, complete interaction custody, prospectively frozen evaluation and strict no-censor decision rules are in place; one narrowed unseen transfer is positive, while the wider development record remains mixed	Seal the full-V3 predictor, open the 100-target panel, execute both matched systems and apply the frozen parity decision

The first forecast is strongly supported by observed exact-solver scaling and repeatable independent reconstruction. The second is supported at the carrier and partial-recurrence levels but still depends on complete joint recurrence. The third is the programme's registered empirical prediction: it is the result the completed system is expected to achieve and the matched blind campaign is designed to decide. The adverse 4APD, 4B19 and 8HJC results and the two partial-recurrence regressions are reasons to demand the full relation and blind protocol, not reasons to erase or reclassify the forecast.

When those operations are complete, this first Zenodo release will receive a new version containing the reconciled paper, complete audit layer, machine archive and final parity adjudication. Zenodo's immutable first-version record will continue to show exactly what was known, adverse and unresolved at this preliminary boundary.

14. Limitations and open frontier

The following boundaries are current:

- the exact finite representation and complete interaction custody are implemented;
- the source-bound coordinates are numerical emission records, not exact SFT equality authority;
- plural protonation and exact local-transfer custody are complete; the held condition is resolved through the joint relation rather than a direct fitted selector;
- complete pair enumeration supplies no physical order by itself;
- the historical partial relation remains plural and non-discriminating, while the later authorised joint relation constructs all six preserving physical axes at the declared fixed-geometry boundary;
- the componentwise partial order has three nondominated states and therefore supplies no unique representative;

- exact recurrence closes the partial result as three all-pair terminal classes, not one fold;
- exact-point exclusion is complete only at point equality and supplies no finite exclusion or exchange scale;
- the historical 17-route census has zero complete survivors and is preserved; the later 256-form joint derivation has one preserving survivor;
- abstract finite order existence does not replace the registered joint physical relation or its required execution;
- the generic transition relation is non-discriminating;
- the generic exact joint frontier solver exists, but the complete fixed-geometry frontier and arbitrary-sequence joint recurrence have not been executed;
- a target-independent partial recurrence exists and closes 3/3 development sequences, but its structural accuracy regresses on 2/3 and it is not the certified full-V3 representative rule;
- no primary 100-target V3 panel has been selected or opened;
- the AlphaFold 3 source environment is verified locally, but restricted model parameters, full registered databases and complete matched execution remain unavailable; and
- this action-specific authorisation covers this updated preliminary Zenodo version and its matching main-branch Git push; it does not authorise a final parity publication.

Completion of the current programme requires complete execution of the six-axis fixed-geometry frontier, the whole-chain torsion–protonation–solvent recurrence, successful development and adverse controls, a sealed 100-target campaign, matched official AlphaFold 3 execution, strict parity adjudication, final paper and archive reconciliation, and Maria Smith's approval. Even that completion would close only this versioned investigation; protein science and lawful extensions would remain open.

15. Conclusion

The V3 programme has converted protein folding from a collection of favourable reconstructions into an explicit dependency and evidence problem. Sequence, condition, torsion words, source-bound whole-chain coordinates, atom pairs, shared-atom correlations, archive verification, metric evaluation and strict parity adjudication now have distinct executable layers. The construction also found and corrected a real carbonyl-plane defect because every pair was retained rather than hidden behind a cut-off.

The resulting scientific status is narrower than a fold claim and stronger than an undocumented failure. The historical partial gate lawfully halted. The later joint-microstate programme constructs all six preserving physical axes, an exact fixed-geometry compiler and a generic certified solver, but its full frontier and whole-chain recurrence are not complete. The adverse V2 generalisation record, adverse 4APD control, non-discriminating recurrence probe, corrected defect, adverse 8HJC transfer, positive narrowed 1JDM transfer and unavailable comparator assets all remain part of the evidence. The generalised partial recurrence closes on all three current development sequences but improves structural TM-score on only one, so its two cross-target regressions also remain active adverse evidence.

This standalone preliminary paper therefore contributes a verified representation, custody, development chronology and failure-semantics spine to the wider SFT dependency tree. It also supplies a sharply bounded positive sequence-only transfer that can be tested further without erasing its adverse predecessor. The observed scaling and independent reconstruction make continued exact closure the most likely programme outcome. The resulting whole-chain system is predicted to reach the frozen generalised blind-parity test, whose outcome will be decided by the complete matched evidence. The next scientific act is to complete and certify the registered joint frontier and whole-chain recurrence, then require it to pass development and adverse controls before sealing the matched campaign. Comparator assets remain a separate required execution dependency. The completed investigation will be published as a new version of this Zenodo record.

Data and code availability

All current code, specifications, audits, source captures, tests and development records are held in the repository under `applications/frontier/v3_computational_proofs/protein_folding/`. This first standalone Zenodo release contains the conceptual paper, scientific audit layer, reproducibility record, machine manifest and checksum ledger. Papers and documentation are licensed CC BY 4.0; code is licensed Apache-2.0. Third-party sources retain their own licences and

provenance. Later complete results will be issued through the same Zenodo concept while preserving this version.

Contribution statement

Maria Smith conceived Smithian Fold Theory, directed the programme and retains scientific authorship and publication authority. OpenAI Codex supplied implementation, verification and editorial assistance under the repository's agent restrictions. Automated assistance did not admit claims, breach a pre-seal target boundary, award scientific authority or authorise publication.

References

1. Smith, M. *From Fold to Life: Biology and Life Sciences Foundational Branch Paper 001*, version 1.0.0. Zenodo DOI 10.5281/zenodo.21630203 (2026).
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3. wwPDB Chemical Component Dictionary. Current source captures and snapshot identities are retained in provenance/wwpdb_atomistic_source_capture_v1.json.
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Machine-identity appendix

Artefact	SHA-256
Workspace manifest	9af6e376ef4454d1f764ef6f175ce36d46510525a8c628ac1d11d0ce674229b6
Publication-integrated full test-suite audit v21	9e42ec2958382e2025081224d1a751b126833602d4ba5bca86e99d349c18d9b5
Full test-suite audit v20	7ca92c353bdf37993825124eb9c8e3dc69717fd6671c18a56219a9a4ce5cbb25
Current scientific gate v20	b9483b91c07d0ae2e70125d48dac11fa5a3a489202e894886db3c0f0e4ac6221
Full test-suite audit v19	4a339ce06f0e7272df93b7fbca530458ab5a29872809724ca8e3f7fcb76989eb
Current scientific gate v19	6798649fb4397a4d0c267f0a774419bec6791f9fe318821fd06e7d1bd5133268
42,173,082-candidate parent registration	4bc05e7d963de968ab66b3a3f3af7746dd46d80961489e51fd70bdaddf6ba3f6
42,173,082-candidate parent execution	ac75105b492dff357646e0af78fbb4b5cc7d95d273056e384697815e79e5b0ec
Independent 42,173,082-candidate reconstruction	6d3f40911179060db0056328821d9e3371be3c58005d2140921e82338a3f0c07
42,173,082-candidate parent archive	355de560d02f8d5973e2af2ce9d21653f3a928a8edf0878d75521f8a9a5dd6f3
Exact high-attraction response census	c5887da957d9d8a5643bf0006ef53c314fe27bd475f7b7a3f77aa699d179073a
Independent high-attraction census reconstruction	8ca0c69afddba9b12e34a233b33d6a33716ffa53042a4bb9a555a5b959e97e98
Arbitrary-sequence joint-response compiler	40384b8087ac24fd12600a2a49ee6bbb758d8c01a60a676feacd64c1ac3e33dc
Independent arbitrary-sequence compiler reconstruction	10f1c2ed65aabc813e6a0d8bf1b1ab34cc9e387b1a47a08cc4a95f62a03ec928
Arbitrary-sequence compiler adverse controls	57fcf174df84d15704327da2b858bae76002c1ce3a520ff789a59738676c8fd1

Artefact	SHA-256
Real non-AGA compiler-to-solver control	950a834e789bd87f8970338f744edc05bb89f12545905db054661d9c98d52ed7
Independent six-candidate solver reconstruction	ee9a62c0e682c6f5b7acae7242cddb2d90e6e3fe90167a78674318788f36b1e5
Complete-sector exact incumbent seed	288d872a338619cd0797fa19407666594e34d16cbca294de471489815052bd04
Independent reconstruction of 22 incumbent certificates	eb92499702e501de382f0bfcbbfa2608dbb0eab55404e2a5e987f5a331601df2
Adverse depth-12 root-pruning census	4de5b26ec9c745f973a4b2775ea7f0a8f893474695014fdb92502411a27d1d94
Independent depth-12 adverse-census reexecution	dad110ece2aa5bdbc3148df649cb517925e14f2d9b2aa7ab65cc4e23c60de5
Adverse degree-strengthened depth-8 census	b3288d18dd0a001224f08af22fa4a44e3ab661f3b609d3ac8d9c1fb798436157
Independent degree-census reexecution	5314221284e315342814d4edfe41597751564ed0d5c048c5841da2931103b42f
Adverse complete 59-stratum census	d81eec214e717e51c59594f2a6f5efead702e10984daa447e61a7cc6e57a409e
Independent 59-stratum reexecution	370e4b86356639ddec6a2527d418386c7cf2fec1f11e08d760bdf4ed78f6c4e6
Full test-suite audit v18	68aae5216eed70d9b765d7b2a5f4791f9ac3b9e61761f0695552868dada7fdad
Current scientific gate v18	bedbcb348216c336343fa6bc82300d6cffe48d2b8205bf4fcf51e6013cd749
3,347,070-candidate parent execution	3a83273dce3a4615070e081a3d95494855af6846272fb6c8eb9ea521fb61148a
Independent parent reconstruction	b8ebfabb7aeeab49bf6eb1d956299ec8c6c75e80d5041021e1ef81bc2be3a110
3,347,070-candidate parent archive	3f6a585364276f58af15c357c2ce8597780d8109b837fcc9079c4b138322c2ac
Full test-suite audit v5	d9edfc3f42c3180f2545ac929780758f4089bfff71a821827fdf8b6379b7d6dad
Current scientific gate v7	1db989772ee69abc5abb8b5f800925e51de5666b8ce26a2fee9e5c22d1aa884b
Current scientific gate v13	b4ad828736d3d4caf1382df768f87be72b1e054092574caa526c04cf1eae8a86
Current joint-microstate status v6	5878ccbfd2e861fae6159c3e79ab642407fe3377e02e8afa3b99ce4f3ae03fbd
Full test-suite audit v13	5211289b2914317049243148c74538119dfee35d1ed798c4965f854b1e74d6ee
Current scientific gate v14	005116be52cf247cfac3d2220727c08851769f1c46a1a012388167956735d8e2
Current joint-microstate status v7	f232963f487f79616550c894c903657b499fe986b17151ff44fef93e301e1143
Full test-suite audit v14	1ce2abf66970f71843f648a3b2038e6d93f7e4fc891a0d55bdbdfbf77a542c8c
Full partitioned execution plan	4441f6659e16ba53b308eb600db30fff11e2ca082ede54f3a2458d9fcbe197b4f
Second real deep-shard audit	3cb24d7c9f1ebba4a57c86c512e7a1b322b08692da5b075ff70551e84e4fe389
Second real deep-shard verification	7d2cbe28ba37507703be25e739404a4411f5562b6823fb591aa3aa6345e1be1f
Third real deep-shard audit	bca34d30d0e08cc0d03c77cee3a08a9a71cc3001aae359b2c0a89c0532e8d
Third real deep-shard verification	2ee5d920e9ab4da6c79249bc1ced31a07f749c402e2e0fb49f38fa7ad5b9824b
Predictor execution gate v2	f38a469790e7759e8e842acd69d2f8e3da4d098ea73401158930eed1d400a67c
Predictor execution gate v2 verification	5c84f77da582717620b3128a07d9737638a1bdfd2a69139cc05b532482431fdd
State-dependent-height route census	3f307fec7ea7de3a22dc4dd97ca1d98e5a5e401cbd52dd35d32350def1a71786
Partial-coordinate recurrence certificate	18736d9df0132c7b0727b36fe4451ed808bda36d2505ae68755b53b96fb971f0
Conformation-order no-height certificate	e3d0ea6bce84888ffba443525c38b954afd6dcb80014d332a37bdd776ebe8fe7
Full-coordinate completion census v2	63221b39e08b9cb5914a8028019e814d243b7116620bd4537b6a8933672ef836
Whole-chain interaction custody specification	078ffbbf757c0b9d2c6bd8cf9810aa9e1340338b74067ecc14d844cd781af9c4
Whole-chain interaction custody audit	5bbac7f0ade84250994cdf2b7190ae3bc25301aa0d7261ac3d110caaa1b01fec
Corrected interaction archive	a3b2563d81374c61418360fdbb43745c15524fa5a79cdf9c05c54efcbcfbb791
Predictor execution-gate registration	49046c64cd753de03f23d6018f34454bc2d4535872211af1800f9f9af04b094f

Artefact	SHA-256
Predictor execution-gate audit	7a96992aec794a49131b1c4242d31412c8f97fc5d8905477ae6e499c804ceb46
Sealed capability-halt result	aa59916dfa007264f6d49a1a156bd538c3d74502bcf05efa1a7d45ecce639753
Physical-coordinate census specification	97a6e1b2175c3818f4892a2d8f85ee3e13ab419a6586a8261ea9484795f5413c
Physical-coordinate census audit	b352134a505e1f58b80185a2f0a6f74eeb1730f56a96b4d08265961aab9d68f6
Canonical campaign evaluator v2	b2c006018d9fea56c04ef7301537475688326d4dbcaae3221cc30ccb8eef4e7d
TM-align executable	02850a90e548bdd7d2c2280124649323efde2c4aa0d8580348e9f00a701420f6
AlphaFold 3 frozen source archive	f81dd985fe4ad0d24a365c47f5740bde904eb95bd7ab6c2ae5a366dcc58df93a
AlphaFold 3 local environment audit	ede7e73ede462963d1db0f20cd9a4c250423457a71cbe76c83329bdbc125651e
8HJC official TM-align evaluation	2603c78b3bf98d78e8774830c4297d14edf47feb3623e8f9481ae0c1a290952b
1JDM official TM-align evaluation	c8e72771ac5069665bfd3c9be677720a7e9a491253fca48e8e8562346f76ab0f

The complete path and identity inventory is retained in the machine archive manifest rather than repeated throughout the conceptual narrative.